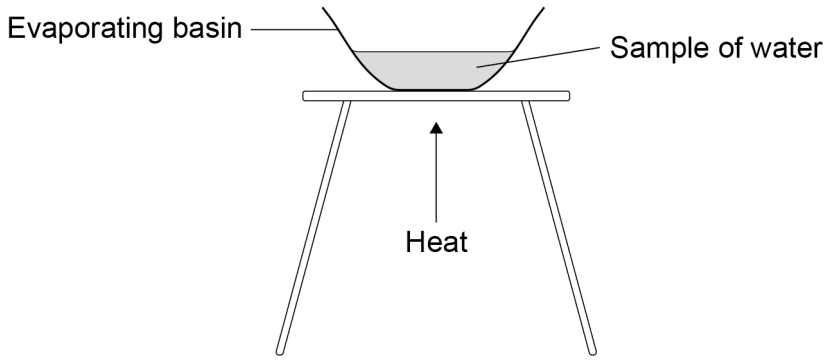
Do not write outside the box

0 3 Some types of water contain dissolved substances.

A student investigated the mass of dissolved solids in distilled water and in sea water.

Figure 5 shows the apparatus.

Figure 5



This is the method used.

1. Weigh an evaporating basin.
2. Add a sample of distilled water to the evaporating basin.
3. Weigh the evaporating basin and the water sample.
4. Heat the water sample.
5. Weigh the evaporating basin and contents.
6. Repeat steps 1 to 5 two more times.
7. Repeat steps 1 to 6 with sea water.

0 3 . 1 Give **one** safety precaution the student should take.

[1 mark]

_____

_____

Do not write outside the box

0 3 . 2 The method used did **not** give valid results.

Which **two** improvements could the student make to give valid results?

[2 marks]

Tick (✓) **two** boxes.

Heat until the mass of the evaporating basin and contents does not change. ☐

Leave some water in the evaporating basin after heating. ☐

Record the results in a table and plot a bar graph. ☐

Use 25 cm^3 of water for each experiment. ☐

Use a beaker instead of an evaporating basin. ☐

0 3 . 3 A different student used a method which gave valid results.

Table 2 shows the results.

Table 2

	Mass of dissolved solids in grams		
Type of water	**Test 1**	**Test 2**	**Test 3**
Distilled water	0.00	0.00	0.00
Sea water	0.31	0.32	0.27

The student concludes that distilled water is pure.

Suggest **one** reason why.

[1 mark]

_____

_____

Do not write outside the box

Tap water is potable.

One step in producing potable water is sterilisation.

0 3 . 4 Why is potable water sterilised?

[1 mark]

Tick (✓) **one** box.

To improve the taste ☐

To kill bacteria ☐

To remove dissolved solids ☐

0 3 . 5 Chlorine is used to sterilise water.

The reaction between chlorine and water is reversible.

Complete the sentence.

[1 mark]

Equilibrium is reached when the forward and reverse reactions

occur at the same _____ .

0 3 . 6 The test for chlorine uses damp litmus paper.

Complete the sentence.

Choose the answer from the box.

[1 mark]

blue	**green**	**white**

When damp litmus paper is put into chlorine gas

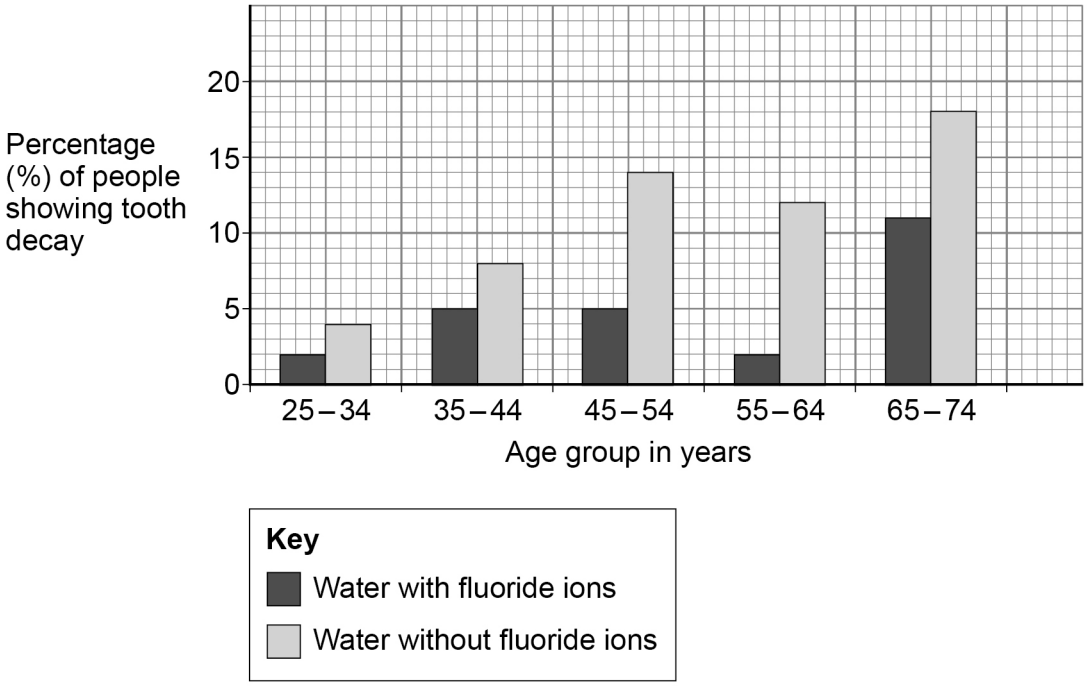
the litmus paper turns _____ .

Do not write outside the box

There is evidence that adding fluoride ions to drinking water reduces tooth decay.

Figure 6 shows the percentage of people showing tooth decay.

Figure 6



0 3 . 7 Compare the effects of adding fluoride ions to drinking water on tooth decay in people between the ages of 25 and 74.

Use data from **Figure 6**.

[4 marks]

Do not write outside the box

0 3 . 8 Student **A** says that water companies should add fluoride ions to all drinking water.

Student **B** disagrees.

Suggest **one** reason why Student **B** may think that fluoride ions should **not** be added to all drinking water.

[1 mark]

_____

_____

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